

EPPS6323 Knowledge Mining

Assignment 2: Prompt exercise

Objective: Design prompts to conduct a structured systematic literature review on data mining and machine learning using multiple AI models (e.g. ChatGPT, Copilot, and Grok 3) and analyze each model's contribution and refining prompts for academic precision and depth.

Exercise Steps:

1. Step 1: Initial Prompt Creation

- Task: Write a baseline prompt to request a structured systematic literature review on data mining and machine learning applications.
- Example Prompt: “Conduct a 2,000-word structured systematic literature review on the applications of data mining and machine learning in real-world domains. Include a methodology section, synthesize key findings, identify trends and gaps, and propose one testable hypothesis. Use an academic tone and emulate systematic review standards.”
- Students submit this prompt to ChatGPT, Copilot, and Grok 3, collecting the raw outputs.

2. Step 2: Analyze Model Responses

- Task: Assess each model's output based on:
 - **Structure:** Did it include a methodology section and follow a systematic review format?
 - **Synthesis:** Were key findings from data mining and machine learning applications well-summarized?
 - **Trends and Gaps:** Did it identify meaningful trends and research gaps?
 - **Hypothesis:** Was the proposed hypothesis testable and relevant?
 - **References:** Are the citations accurate (check using Google Scholar or Semantic Scholar)
- Students document strengths and weaknesses. For example, ChatGPT might provide broad coverage, Copilot might excel at concise structuring, and I (Grok 3) might offer a unique perspective on emerging trends.

3. Step 3: Refine the Prompt

- Task: Revise the prompt to address deficiencies in each model's response, creating three tailored prompts—one for ChatGPT, one for Copilot, and one for Grok 3.
- Example Refined Prompt for Grok 3: “Imagine you're a data scientist conducting a 2,000-word systematic literature review on how data mining and machine learning are applied in domains like healthcare, finance, and education. Outline a clear methodology, synthesize key findings with fresh insights, highlight trends and gaps, and propose one bold, testable hypothesis. Maintain a rigorous academic tone.”

- Students test these refined prompts and compare the improved outputs.
- 4. **Step 4: Cross-Model Collaboration**
 - Task: Integrate the best elements from each model's output into a final systematic review. Write a new prompt for the student's preferred model (e.g., Grok 3) to synthesize the results.
 - Example Synthesis Prompt: "Using these drafts from three AI models [paste outputs], produce a 2,000-word structured systematic literature review on data mining and machine learning applications. Combine the strongest methodology, findings, trends, gaps, and hypothesis into a cohesive, academically sound document."
 - Students submit their final review and justify their synthesis decisions.
- 5. **Step 5: Reflection**
 - Task: Write a reflection answering:
 - How did each model approach the systematic review differently?
 - Which prompt refinements yielded the best results for each model?
 - What did you learn about leveraging AI for structured academic reviews?